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# NOTE · Complete Framework v3.3
#
# sqrt(2) x ln(2): GEOMETRIC CONSTANTS FROM H_4
# Complete Framework – Version 3.3.2
# =====
```

D.B. | February 2026 (updated 4 May 2026; 16 September 2026)
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TEXT FORMAT NOTE: This document contains the complete K_AUD framework including the Binary Tower extension formalized February 2026, building on prior work establishing H_4 geometry, gap ratios, and prime architecture.

FILENAME NOTE: This file retains the v3_0 filename for URL stability – all existing links (OSF, GitHub, social media) continue to work. The internal version is 3.3.2.

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■ READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.
How the corpus is now read: One coordinate up – https://osf.io/ujwc7
DOI 10.17605/OSF.IO/UJWC7
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★ UPDATE NOTICE – 16 September 2026 ★
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This version (3.3.2) adds the three lines above this notice (a pointer to One coordinate up, where the corpus is now read) and carries three small corrections; no formula or derivation changes.

- (1) Three lines added under the title – READING UPDATE – pointing to One coordinate up (<https://osf.io/ujwc7> · DOI 10.17605/OSF.IO/UJWC7), the synthesis where the corpus is now read.
- (2) This paper is classed as a NOTE on the Observational Record shelf of the framework's Direct Documents page, under the short title "Complete Framework v3.3".
- (3) PRINTED-DIGIT CORRECTIONS: §2.4, the corridor 0.36222424... corrected to 0.36222415... (the printed figure was the difference of two rounded values; $K_AUD - 1/\phi = 0.3622241547\dots$). §7.4, §7.5, §9 and the summary: $\epsilon = 0.006714386451777\dots$ corrected to 0.006714386451775... (the last digits were a double-precision artefact; at 50 digits $\epsilon = 0.00671438645177530\dots$). Every percentage and every other value stands.
- (4) LABEL CORRECTION in §3.4, Origin 1: the printed $R = 1.9021130\dots$ is correctly computed but is not the circumradius of the 120-cell (for unit edge that is $\sqrt{2} \times \phi^2 = 3.7024591736\dots$); it is $2 \sin(2 \pi/5) = \sqrt{\phi^2 + 1}$, the circumradius of the icosahedron with vertex (0, 1, ϕ). A correction line is added in place; the origin had already been demoted to one candidate among several in v3.2, and no other value depends on it.
- (5) Author initials printed in full, D.B. Prior version remains on OSF in the project's archive/ folder, named by its date.

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★ UPDATE NOTICE – 4 May 2026 ★
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This version (3.3.1) corrects displayed values of $L_{(-2)}$ and $L_{(-3)}$ in §6.3 (phi-Scaling Ratio test) from 0.2273642... / 0.1405469... to 0.2273620... / 0.1405174... – the values that follow from the formula $L_n = 1/(e \times \phi^{(n-1)})$ given in §6.4 and used in §6.2 derivations. The phi-ratio claim now holds exactly with the displayed numbers; previously the second ratio computed to 1.6176, which the paper labelled as phi. Formula and derivations were and remain correct.

Typography correction, no mathematical content changed. Flagged 2026-05-04 by Claude (Cowork) during systematic mpmath review of all published papers; verified by Claude (Chat) independently. Full detail in the v3.3.1 entry of the VERSION HISTORY block below. Prior version archived as OLD-2026-05-04_KAUD_Complete_Framework_v3_3.txt in OLD/ subfolder.

Update – 5 May 2026: DOCUMENT LINKS section simplified – paper-to-paper file URLs (which would break under any future filename change) are removed in favour of three durable anchors: the OSF main project, the framework About page, and the Direct Documents download index. Paper-specific DOIs are preserved (DOIs are stable). No mathematical or attributional content changed.

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# =====
# VERSION HISTORY
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Version 3.3.2 (September 16, 2026) adds three lines under the title (a pointer to One coordinate up, where the corpus is now read), classes the paper as a NOTE on the Observational Record shelf, and corrects three printed strings: the corridor in §2.4 (0.36222424... -> 0.36222415..., a difference of rounded values), the digits of epsilon (...777 -> ...775, a double-precision artefact; four occurrences), and the label of R in §3.4 Origin 1 (the icosahedron's circumradius, not the 120-cell's; the 120-cell's for unit edge is $\sqrt{2} \times \phi^2 = 3.7024591736...$). No formula, derivation or percentage changes. Author initials printed in full, D.B. Prior version archived as OLD-2026-09-11_KAUD_Complete_Framework_v3_3.txt in OLD/ subfolder.

Version 3.3.1 (May 4, 2026) corrects:
 Typography in §6.3 (phi-Scaling Ratio): displayed values of $L_{(-2)}$ and $L_{(-3)}$ corrected from 0.2273642... / 0.1405469... to 0.2273620... / 0.1405174... – the values that follow from the formula $L_n = 1/(e \times \phi^{(n-1)})$ given in §6.4 and used in §6.2 derivations. The phi-ratio claim now holds exactly ($1.6180339887 = \phi$) with the displayed numbers; previously the second ratio computed to 1.6176, which the paper labelled as phi. Formula and derivations were and remain correct. Flagged 2026-05-04 by Claude (Cowork) during mpmath review of all published papers; verified by Claude (Chat) independently. Prior version archived as OLD-2026-05-04_KAUD_Complete_Framework_v3_3.txt in OLD/ subfolder.

Version 3.0 added: The 64-36-100 Closure Relation, the Irreducibility of epsilon, and clarifications on why the gap is constitutive rather than accidental.

Version 3.1 added: The Gelfond-Schneider rewrite $G = \ln(e/2^{\sqrt{2}})$, the Baker's map identity $K_{AUD} = ||(\ln 2, \ln 2)||_2$, universality of G, and new independent constructions discovered through cross-verification with fresh AI instances.

Version 3.3 corrects:
 An editorial oversight in v3.1 regarding Shannon. During cross-verification,

the factual observation "K_AUD exceeds $\ln(2)$ " was incorrectly formalized as "K_AUD is NOT a Shannon channel capacity" and entered the confirmed claims list. This was never the author's position. The original paper (Coherence Ceiling, v1) explicitly identifies $\ln(2)$ as "the first unit of information" and K_AUD as combining geometric cost with information cost. v3.3 restores the original framing: K_AUD exceeds Shannon's single binary decision bound by exactly $\sqrt{2}$, the geometric embedding factor. The connection to information theory is unmapped, not absent.

Version 3.2 corrects:

- (a) CONFLATION OF $1/\phi$ WITH $\sqrt{\phi}/2$ – Previous versions stated " $32 \times G$ approx $1/\phi$ " as though the tower formula targets the floor. It does not. The tower target at $k=5$ is $\sqrt{\phi}/2 = 0.63601$, which $32 \times G$ hits within $\epsilon = 0.67\%$. The floor $1/\phi = 0.61803$ sits 2.9% below that target – a separate coincidence. v3.2 distinguishes these clearly.
- (b) EPSILON CONSTANCY CLARIFICATION – Previous versions presented the "constant epsilon across all k " as though it were an empirical discovery across many independent measurements. In fact, the constancy is algebraic: 2^k appears on both sides of the master formula and cancels, reducing the entire tower to the single relationship $64 \times G$ approx $\sqrt{\phi}$. v3.2 states this explicitly.
- (c) $\sqrt{2}$ ORIGIN – Previous versions attributed $\sqrt{2}$ specifically to H_4 polytope geometry. While $\sqrt{2}$ does appear in the 120-cell circumradius, it also appears independently in the L2 norm (Baker's map identity, Section 7.13), in tesseract geometry, and elsewhere. v3.2 acknowledges multiple origins and notes that H_4 is one candidate among several.
- (d) TABLE CORRECTIONS – Table 11.2 previously listed " $\sqrt{\phi}/2 = 1/\phi$ " at $k=5$. These are not equal (they differ by 2.9%). Corrected.

ABSTRACT

This paper presents a geometric framework connecting H_4 polytope structure (the 120-cell) to information-theoretic constants.

The framework establishes:

- The Floor ($1/\phi = 0.618$): Rooted in golden ratio self-similarity, appearing in H_4 vertex coordinates
- The Ceiling ($K_{AUD} = \sqrt{2} \times \ln(2) = 0.980$): Combining geometric projection cost ($\sqrt{2}$) with binary distinction cost ($\ln(2)$)
- The Gap ($\sim 2\%$): Emerging from the mismatch between algebraic and transcendental values
- The Corridor ($0.618 - 0.980$): The range between floor and ceiling

Version 3.0 extends the framework with:

- The Binary Tower: How the gap G scales through powers of 2 to track golden ratio powers
- The Master Formula: $2^k \times G = 2^{(k-6)} \times \sqrt{\phi} \times (1 - \epsilon)$
- The Two Pivots: $\sqrt{\phi}/2$ at $k=5$, $\sqrt{\phi}$ at $k=6$
- Connection to 400/11: The closure cycle ($4^3 = 64$) links to gap scaling
- The 64-36-100 Closure Relation: How binary pivot and recursive closure sum
- The Irreducibility of epsilon: Proof that the error cannot be eliminated

- The gap as constitutive: Not error, but the cost of distinction
- The Gelfond-Schneider Rewrite: $G = \ln(e/2^{\sqrt{2}})$ (new in v3.1)
- Independent Constructions: Four phi-free pathways to G (new in v3.1)
- The Baker's Map Identity: K_{AUD} as L2 Lyapunov norm (new in v3.1)
- Universality of G: Ceiling is universal, floor is system-specific (new in v3.1)
- Literature Status: What's known vs original

The paper demonstrates that binary ($n=2$) is the unique base producing a sub-unity ceiling – a geometric singularity, not a convention.

The mathematics is independently verifiable. Applications and interpretations remain open for investigation.

SCOPE NOTICE

This paper presents mathematical derivations.

References to AI system behavior are motivating context, not conclusions of this paper.

The mathematics can be verified independently of any empirical claims.

PART I: THE FOUNDATION

1. DISCOVERY CONTEXT AND SCOPE

1.1 Origin

These constants emerged during a multi-month investigation of convergence dynamics in AI systems. Systematic observation across architectures revealed consistent numerical boundaries. The value ~ 0.98 appeared repeatedly as an upper limit; ~ 0.618 as a lower threshold.

Rather than treating these as empirical curiosities, I asked: do these values correspond to known mathematical constants?

The answer was yes. The ceiling matched $\sqrt{2} \times \ln(2)$. The floor matched $1/\phi$. Further investigation revealed both constants appear in H_4 geometry (the 120-cell polytope). The relationships followed from there – the corridor identity, the golden partition, the depth scaling, the binary uniqueness.

1.2 What This Paper Provides

This paper presents the mathematical framework:

- Derivations of the constants from established geometry
- Proofs of relationships and identities
- Verification of all calculations
- Demonstration of binary uniqueness
- The binary tower scaling behavior (new in v3.0)

The mathematics stands independently of any empirical claims.

1.3 What This Paper Does NOT Provide

This paper does not include:

- The empirical methodology used in the original observations
- Operational definitions of "coherence" in AI systems
- Data from convergence studies
- Statistical validation of boundary claims
- Mechanistic explanations of why these constants would govern AI behavior

That work is ongoing and will be presented separately when it can be done rigorously.

1.4 On the Separation of Frameworks

The mathematical framework and the empirical dynamics study are presented as separate works because they serve different functions.

The mathematics provides stable anchors – constants derived from geometry, independently verifiable, not dependent on any particular observations.

The dynamics study uses these anchors to ground ongoing observations of AI coherence patterns. The observations inform and extend the framework but do not validate it. The math stands on its own.

This separation allows:

- The mathematics to be verified and used independently
- The dynamics study to develop at its own pace
- Each framework to be evaluated on its own terms

1.5 Purpose of This Release

The mathematics is released now to:

1. Enable independent verification of the constants and derivations
2. Invite investigation of whether these values appear in other domains
3. Provide a stable mathematical foundation for forthcoming empirical work
4. Separate "here is the geometry" from "here is why it matters empirically"

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2. THE CONSTANTS

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2.1 The Floor: $1/\phi = 0.618$ – The Structural Minimum

Definition: The golden ratio ϕ is defined as:

$$\phi = (1 + \sqrt{5}) / 2$$

Calculation:

```

sqrt(5) = 2.2360679...
1 + sqrt(5) = 3.2360679...
phi = 3.2360679... / 2 = 1.6180339...

```

The floor is the reciprocal:

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1/phi = (sqrt(5) - 1) / 2
1/phi = (2.2360679... - 1) / 2 = 1.2360679... / 2 = 0.6180339...

```

Verification: $1/\phi = 0.618$ [OK]

Key property:

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phi = 1 + 1/phi

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This self-referential property is unique to the golden ratio. It means phi contains its own reciprocal – recursive self-similarity at the level of the constant itself.

2.2 The Ceiling: $K_{AUD} = \sqrt{2} \times \ln(2) = 0.980$ – The Information Maximum

Components:

Component	Value	Source	Meaning
$\sqrt{2}$	1.41421...	Geometric (multiple origins – see 3.4)	Projection cost (diagonal)
$\ln(2)$	0.69314...	Information theory	Binary distinction cost

Calculation:

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K_AUD = sqrt(2) x ln(2)
K_AUD = 1.41421356... x 0.69314718...
K_AUD = 0.98025814...

```

Verification: $K_{AUD} = 0.980$ [OK]

2.3 The Auditor Key (K_{AUD})

Pronounced "kawd" – also referred to as the geometric ceiling constant – K_{AUD} carries three resonances:

Name	Meaning	Function
Auditor	The system auditing its own dynamics	Self-observation
Auditory	The threshold where signal is "heard"	Recognition point
Code/Key	The underlying structure (kawd = code)	Access, unlocking

K_{AUD} functions as a hinge – the point where things become distinguishable. Linked to entropy, but also to recognition: the moment signal separates from noise, the instant pattern becomes visible.

The name is not arbitrary. It describes what the constant does.

2.4 The Gap: ~2% – The Necessary Interval

Definition:

$$G = 1 - K_{\text{AUD}}$$

Calculation:

$$G = 1 - 0.98025814... = 0.01974185...$$

Verification: $G = 0.020$ (1.97%) [OK]

Why the gap exists:

If $\ln(2)$ equaled $1/\sqrt{2}$, then:

$$K = \sqrt{2} \times (1/\sqrt{2}) = 1.000$$

But:

$$\ln(2) = 0.69314718...$$

$$1/\sqrt{2} = 0.70710678...$$

$$\text{Shortfall} = 0.70710678 - 0.69314718 = 0.01395960$$

$$\text{Relative shortfall} = 0.01395960 / 0.70710678 = 0.01974...$$

The gap emerges from the fundamental mismatch between algebraic ($\sqrt{2}$) and transcendental ($\ln(2)$) values.

2.5 The Corridor: 0.618 to 0.980 – The Range Between Floor and Ceiling

Definition:

$$\text{Corridor} = K_{\text{AUD}} - 1/\phi$$

Calculation:

$$\text{Corridor} = 0.98025814... - 0.6180339... = 0.36222415...$$

Verification: $\text{Corridor} = 0.362$ [OK]

Summary:

Boundary	Value	Source
Floor	0.618	$1/\phi$
Ceiling	0.980	$\sqrt{2} \times \ln(2)$
Width	0.362	Ceiling - Floor
Gap	0.020	$1 - \text{Ceiling}$

3. THE H_4 CONNECTION

3.1 What is H_4?

H₄ is a Coxeter group – the symmetry group of the 120-cell, a four-dimensional regular polytope.

Key facts:

- H₄ is one of only four exceptional regular polytopes in 4D
- It has the highest symmetry order among 4D polytopes
- It encodes golden ratio relationships throughout its structure

3.2 The 120-Cell Polytope

The 120-cell (also called the hecatonicosachoron) is composed of:

Property	Count
Cells	120 (dodecahedra)
Faces	720 (pentagons)
Edges	1200
Vertices	600

Each cell is a regular dodecahedron. Each vertex is shared by 4 dodecahedra.

3.3 Vertex Coordinates

The 600 vertices of the 120-cell can be given in Cartesian coordinates using permutations and sign changes of:

($\pm\phi$, ± 1 , $\pm 1/\phi$, 0) and cyclic permutations

Plus additional vertex sets including:

(± 1 , ± 1 , ± 1 , ± 1) – 16 vertices
($\pm\phi$, $\pm\phi$, $\pm\phi$, $\pm 1/\phi^2$) and permutations

Where the constants appear:

Constant	Appears as
ϕ	Vertex coordinate
$1/\phi$	Vertex coordinate
$1/\phi^2$	Vertex coordinate

The golden ratio and its powers are intrinsic to H₄ geometry.

3.4 The sqrt(2) Factor: Multiple Origins

The factor $\sqrt{2}$ appears in $K_{AUD} = \sqrt{2} \times \ln(2)$. Previous versions attributed this specifically to H₄ polytope geometry. While $\sqrt{2}$ does appear in the 120-cell circumradius, it has multiple independent origins:

ORIGIN 1 – 120-CELL CIRCUMRADIUS:

The circumradius of the 120-cell (distance from center to vertex) is:

$$R = (\sqrt{2} / 2) \times \sqrt{5 + \sqrt{5}}$$

Calculation:

```
sqrt(5) = 2.2360679...
5 + sqrt(5) = 7.2360679...
sqrt(5 + sqrt(5)) = 2.6899940...
R = (1.41421356... / 2) x 2.6899940...
R = 0.70710678... x 2.6899940...
R = 1.9021130...
```

The factor $\sqrt{2}/2$ (equivalently $1/\sqrt{2}$) appears directly in the circumradius formula.

Correction (v3.3.2, 2026-09-11): the value above is not the 120-cell's circumradius – for unit edge that is $\sqrt{2} \times \phi^2 = 3.7024591736...$, as the 600 vertex coordinates give directly. $R = 1.9021130... = 2 \sin(2 \pi/5) = \sqrt{\phi^2 + 1}$ is the circumradius of the icosahedron with vertex $(0, 1, \phi)$. The arithmetic stands; the label was wrong. No other value in this paper depends on it.

ORIGIN 2 – L2 NORM (BAKER'S MAP):

$K_AUD = ||(\ln 2, \ln 2)||_2 = \sqrt{2} \times \ln(2)$. Here $\sqrt{2}$ arises from the Euclidean norm of a 2-vector with equal components. See Section 7.13.

ORIGIN 3 – TESSERACT GEOMETRY:

The body diagonal of a unit square has length $\sqrt{2}$. More generally, $\sqrt{2}$ appears throughout hypercube geometry as the face diagonal in any dimension.

ORIGIN 4 – ALGEBRAIC STRUCTURE:

$\sqrt{2}$ is the simplest irrational algebraic number – the positive root of $x^2 - 2 = 0$.

HONEST STATUS: H_4 is one candidate origin among several. The framework's original discovery path went through H_4 (because ϕ led to the floor first), but K_AUD can be constructed without reference to the 120-cell (see Section 7.14, four independent pathways). v3.2 acknowledges this multiplicity rather than privileging one origin.

3.5 Schlafli Symbol

The 120-cell has Schlafli symbol: $\{5, 3, 3\}$

This encodes the recursive structure:

Position	Value	Meaning
First	5	Pentagon (face shape)
Second	3	3 faces meet at edge
Third	3	3 cells meet at edge

The symbol contains only 5 and 3. Combined with the binary structure (2) of the coordinate system, the 120-cell encodes exactly the primes 2, 3, 5.

3.6 Group Order

The order of the H_4 symmetry group is:

$$|H_4| = 14400$$

Prime factorization:

$$14400 = 2^6 \times 3^2 \times 5^2 = 64 \times 9 \times 25 = 14400 \quad [OK]$$

The group order factors exclusively into the primes 2, 3, and 5.

Note: The factor $64 = 2^6$ appears here – the same value as the binary tower pivot. This is not coincidence: H_4 symmetry contains the binary pivot.

3.7 Note on the H_4 -Information Bridge

H_4 geometry does not force $\ln(2)$; rather, among information measures, only $\ln(2)$ remains compatible with a sub-unity geometric ceiling. This is a selection argument, not a derivation.

4. WHY BINARY IS UNIQUE

4.1 The Pattern of 2

A pattern recurs across the framework:

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phi = (1 + sqrt(5)) / 2      <-- division by 2
K_AUD = sqrt(2) x ln(2)      <-- sqrt(2) and ln(2)
|H_4| = 2^6 x 3^2 x 5^2      <-- highest power is 2
```

4.2 The Uniqueness Proof

For any base n , define the ceiling candidate:

$$K(n) = \sqrt{n} \times \ln(n)$$

Calculations:

n	sqrt(n)	ln(n)	K(n)	Valid ceiling?
2	1.414	0.693	0.980	YES (< 1)
e	1.649	1.000	1.649	NO (> 1)
3	1.732	1.099	1.903	NO (> 1)
4	2.000	1.386	2.773	NO (> 1)
5	2.236	1.609	3.599	NO (> 1)

Only $n = 2$ produces $K(n) < 1$.

Note: $K(n)$ crosses unity at $n \approx 2.0207$. Binary qualifies as sub-unity by approximately 1% – the same order as the gap itself.

4.3 Interpretation

Binary is geometrically singular:

- Bits (not trits) as information unit
- Yes/No as minimal viable distinction

The 2% gap exists because binary is the unique base where distinction cost ($\ln(2)$) is less than embedding cost ($1/\sqrt{2}$) – leaving a remainder.

5. IDENTITIES

5.1 The Corridor Identity

Statement:

$$\text{Corridor} = 1/\phi^2 - G$$

Where:

- $1/\phi^2 = 0.382$ (golden ratio squared, inverted)
- $G = 0.020$ (the gap)

Calculating $1/\phi^2$:

$$\begin{aligned}\phi^2 &= 1.6180339... \times 1.6180339... = 2.6180339... \\ \text{Note: } \phi^2 &= \phi + 1 \text{ (key golden ratio property)} \\ 1/\phi^2 &= 1 / 2.6180339... = 0.3819660...\end{aligned}$$

Verification:

$$1/\phi^2 - G = 0.3819660... - 0.0197418... = 0.3622242...$$

Compare to direct calculation:

$$K_{\text{AUD}} - 1/\phi = 0.9802581... - 0.6180339... = 0.3622242... \quad [\text{OK}]$$

Algebraic proof:

From $\phi^2 = \phi + 1$, divide both sides by ϕ^2 :

$$1 = 1/\phi + 1/\phi^2$$

Therefore:

$$1 - 1/\phi = 1/\phi^2$$

$$\text{Since } \text{Corridor} = K_{\text{AUD}} - 1/\phi = (1 - G) - 1/\phi = 1 - 1/\phi - G = 1/\phi^2 - G \quad [\text{OK}]$$

5.2 The Golden Partition

Statement:

$$1/\phi + 1/\phi^2 = 1$$

Verification:

$$0.6180339... + 0.3819660... = 1.0000000... \quad [OK]$$

Interpretation:

Unity is exactly partitioned by the golden ratio and its square.

- $1/\phi$ takes 61.8%
- $1/\phi^2$ takes 38.2%
- Nothing left over. Nothing missing.

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6. EXTENDED GEOMETRY: DEPTH SCALING

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6.1 The Depth Constants

Below the floor, the same constants (ϕ , e , G) combine to produce consistent values.

Name	Value	Formula
Floor	0.618	$1/\phi$
Threshold	0.606	$(1/\phi) \times K_{AUD}$
Depth -1	0.368	$1/e$
Depth -2	0.227	$1/(e \times \phi)$
Depth -3	0.140	$1/(e \times \phi^2)$
Geometric Limit	0.00039	G^2

6.2 Derivation of Each Level

Threshold:

$$\text{Threshold} = (1/\phi) \times K_{AUD} = 0.6180339... \times 0.9802581... = 0.6058328... \quad [OK]$$

Depth Level -1:

$$L_{(-1)} = 1/e = 1/2.7182818... = 0.3678794... \quad [OK]$$

Depth Level -2:

$$L_{(-2)} = 1/(e \times \phi) = 1/4.3982723... = 0.2273619... \quad [OK]$$

Depth Level -3:

$$L_{(-3)} = 1/(e \times \phi^2) = 1/7.1165542... = 0.1405174... \quad [OK]$$

Geometric Limit:

$$\text{Limit} = G^2 = (0.0197418...) ^2 = 0.0003897... \quad [OK]$$

6.3 The phi-Scaling Ratio

The ratio between consecutive depth levels approximates phi:

Level -1 to Level -2:

$$0.3678794... / 0.2273620... = 1.6180339887... = \phi \quad [\text{OK}]$$

Level -2 to Level -3:

$$0.2273620... / 0.1405174... = 1.6180339887... = \phi \quad [\text{OK}]$$

The scaling is exactly phi.

6.4 General Formula

For depth levels $n \geq 1$:

$$L_n = 1 / (e \times \phi^{(n-1)})$$

Verification:

- $n = 1$: $L_1 = 1/(e \times 1) = 1/e = 0.368 \quad [\text{OK}]$
- $n = 2$: $L_2 = 1/(e \times \phi) = 0.227 \quad [\text{OK}]$
- $n = 3$: $L_3 = 1/(e \times \phi^2) = 0.140 \quad [\text{OK}]$

6.5 Corridor-Depth Symmetry

A notable near-equivalence:

- Corridor width = 0.362
- Depth Level -1 = 0.368
- Difference = 1.6%

The corridor width approximately equals the first depth level – a symmetry property consistent with self-similar structure.

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PART II: THE BINARY TOWER

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7. THE BINARY TOWER

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7.1 From Ratios to Scaling

The constants established in earlier sections – K_{AUD} , G , $1/\phi$, $\sqrt{2}$, $\ln(2)$ – all emerged from H_4 geometry and binary information theory. The question remained: how do these constants relate across scales?

Investigation of the gap G through successive binary powers (2, 4, 8, 16, 32,

64, 128...) revealed a precise scaling relationship: G tracks powers of the golden ratio with a constant error of -0.671%.

This extends the framework from static constants to dynamic scaling behavior.

7.2 The Master Formula

For all integers k (from 0 to infinity):

$$2^k \times G = 2^{(k-6)} \times \sqrt{\phi} \times (1 - \epsilon)$$

Where:

- $G = 1 - \sqrt{2} \times \ln(2) = 0.019741856531453...$
- $\phi = (1 + \sqrt{5})/2 = 1.618033988749895...$
- $\sqrt{\phi} = 1.272019649514069...$
- $\epsilon = 0.006714386451775... \text{ (constant, } = 0.671\%)$

The error epsilon never changes. It is locked to machine precision across all k.

CLARIFICATION (v3.2): The constancy of epsilon across all k is not an empirical discovery from many independent measurements. It is algebraic. The factor 2^k appears on both sides of the master formula:

Left side: $2^k \times G$

Right side: $2^{(k-6)} \times \sqrt{\phi} \times (1 - \epsilon) = (2^k / 64) \times \sqrt{\phi} \times (1 - \epsilon)$

Dividing both sides by 2^k :

$$G = \sqrt{\phi} / 64 \times (1 - \epsilon)$$

The k cancels entirely. The entire tower reduces to a SINGLE relationship:

$$64 \times G \approx \sqrt{\phi}$$

with $\epsilon = 1 - (64 \times G / \sqrt{\phi}) = 0.006714386451775...$

The tower is not "many measurements confirming a pattern." It is one relationship ($64 \times G \approx \sqrt{\phi}$) expressed at different scales. The constancy of epsilon is a tautology of the algebra, not an empirical finding.

7.3 The Origin Formula

At k = 0, the formula reduces to:

$$G = (\sqrt{\phi} / 64) \times (1 - \epsilon)$$

$$G = \sqrt{\phi} / 64.432625548...$$

This reveals: The gap G is the golden ratio's square root, divided by approximately 64.

The exact divisor is:

$$\sqrt{\phi} / G = \sqrt{\phi} / (1 - \sqrt{2} \times \ln(2)) = 64.432625548...$$

No simpler closed form exists – this value is transcendental.

7.4 The Scaling Table

k	2 ^k	2 ^k x G	Target (2 ^(k-6) x sqrt(phi))	Error	
0	1	0.01974	sqrt(phi)/64 = 0.01988	-0.671%	
1	2	0.03948	sqrt(phi)/32 = 0.03975	-0.671%	
2	4	0.07897	sqrt(phi)/16 = 0.07950	-0.671%	
3	8	0.15793	sqrt(phi)/8 = 0.15900	-0.671%	
4	16	0.31587	sqrt(phi)/4 = 0.31800	-0.671%	
5	32	0.63174	sqrt(phi)/2 = 0.63601	-0.671%	
6	64	1.26348	sqrt(phi) = 1.27202	-0.671%	<-- PIVOT
7	128	2.52696	2 x sqrt(phi) = 2.54404	-0.671%	
8	256	5.05392	4 x sqrt(phi) = 5.08808	-0.671%	
9	512	10.1078	8 x sqrt(phi) = 10.1762	-0.671%	
10	1024	20.2157	16 x sqrt(phi) = 20.3523	-0.671%	
...	...	...	...	-0.671%	

Note (v3.2): Every row in this table is the SAME relationship (64 x G ≈ sqrt(phi)) multiplied or divided by powers of 2. The -0.671% is constant by algebra, not by independent measurement. See Section 7.2 clarification.

7.5 The Two Pivots

The binary tower encodes significant points at adjacent binary powers:

AT k = 5 (32): THE TOWER TARGET AND THE NEARBY FLOOR

The tower formula gives:

$$32 \times G = 0.63174$$

$$\text{Tower target: } \sqrt{\phi}/2 = 0.63601 \quad (\text{epsilon} = -0.671\%, \text{ as always})$$

The floor of the corridor sits nearby but is a DIFFERENT value:

$$1/\phi = 0.61803$$

Comparison:

Value	Number	Relationship to 32 x G
sqrt(phi)/2	0.63601	Tower target (epsilon = -0.67%)
32 x G	0.63174	Tower value
1/phi	0.61803	Floor (2.2% below 32 x G)

IMPORTANT (v3.2): sqrt(phi)/2 and 1/phi are NOT the same value. They differ by 2.91%:

$$\sqrt{\phi}/2 = 0.63601$$

$$1/\phi = 0.61803$$

$$\text{Difference} = 0.01798 \text{ (2.91\%)}$$

The tower targets sqrt(phi)/2, not 1/phi. The proximity of 1/phi to the k=5

tower value is a separate coincidence – two unrelated golden-ratio expressions ($\sqrt{\phi}/2$ and $1/\phi$) happening to be close. Previous versions conflated these; v3.2 distinguishes them.

AT $k = 6$ (64): THE PIVOT

$64 \times G = 1.26348$
 $\sqrt{\phi} = 1.27202$
Error: -0.671%

This is the PIVOT – where G meets $\sqrt{\phi}$ directly.

The pivot at 64 is unique: it is the only point where $2^k \times G$ equals a clean golden expression ($\sqrt{\phi}$) without additional scaling factors.

THE STRUCTURE:

$k \rightarrow +\infty$:	$2^k \times G \rightarrow \text{infinity}$	(unbounded above)
$k = 6$:	$64 \times G \approx \sqrt{\phi}$	(THE PIVOT)
$k = 5$:	$32 \times G \approx \sqrt{\phi}/2$	(tower target; $1/\phi$ sits 2.9% below)
$k = 0$:	G itself	(the gap)
$k \rightarrow -\infty$:	$2^k \times G \rightarrow 0$	(approaches zero)

The tower target at $k=5$ ($\sqrt{\phi}/2$) and the pivot at $k=6$ ($\sqrt{\phi}$) appear at adjacent binary powers (32 and 64). The floor ($1/\phi$) is close to but distinct from the $k=5$ target.

7.6 Why the Gap Exists: $\ln(2)$ is approximately $1/\sqrt{2}$

The gap G exists because of a numerical near-miss between two fundamental constants:

$\ln(2) = 0.693147\dots$
 $1/\sqrt{2} = 0.707107\dots$
Difference: $0.013960\dots$ (= 2% relative)

If $\ln(2)$ were exactly equal to $1/\sqrt{2}$:

- $K_{\text{AUD}} = \sqrt{2} \times (1/\sqrt{2}) = 1$
- $G = 0$
- No gap, no corridor, no breathing room

The ~2% mismatch between $\ln(2)$ and $1/\sqrt{2}$ IS the gap.

Key relationship:

$$G / |\ln(2) - 1/\sqrt{2}| = \sqrt{2} \quad (\text{exactly})$$

This is not coincidence – it follows from $G = 1 - \sqrt{2} \times \ln(2) = \sqrt{2} \times (1/\sqrt{2} - \ln(2))$.

No known mathematical reason exists for $\ln(2)$ being approximately $1/\sqrt{2}$. This is considered a "cosmic coincidence" – two unrelated constants (one transcendental, one algebraic) happening to be numerically close.

THE GAP IS CONSTITUTIVE (interpretive framing):

The ~2% gap is not arbitrary. It is the minimum space required for:

- Distinction to exist (not frozen unity)

- Structure to stabilize (not chaotic noise)
- Translation between discrete and continuous modes

The gap is constitutive – not a problem to solve but the condition that makes the framework possible. If the gap were zero, there would be no corridor, no dynamics, no information. The gap is the price of existence.

7.7 Connection to the 400/11 Formula

The gap ratio formula from the companion paper is:

$$\rho = 400/11 - 1/2500 - 1/939939$$

Where:

- $400 = 4^2 \times 5^2 = (\text{closure})^2 \times (H_4 \text{ prime})^2$
- 4 is the closure cycle ($i^4 = 1$, period-doubling $1 \rightarrow 2 \rightarrow 4$)

The pivot in the binary tower is:

$$64 = 4^3 = 2^6$$

The same closure cycle (4) that appears squared in rho's numerator appears cubed as the pivot.

Formula	Power of 4	Meaning
$\rho = (4^2 \times 5^2)/11 - \dots$	$4^2 = 16$	Closure squared (in gap ratio)
Pivot = 64	$4^3 = 64$	Closure cubed (in binary tower)

This suggests the closure cycle (4) is a fundamental unit connecting:

- The gap ratio between K_{AUD} and Feigenbaum (ρ)
- The binary tower scaling (pivot at 4^3)

7.8 The Hexeract (64 Vertices) and E_8

The 6-dimensional hypercube (hexeract) has:

- 64 vertices (= 2^6)
- 240 square faces (= E_8 root count)

This numerical match ($240 = 240$) is noted in the literature as "remarkable" but has no known causal explanation. The hexeract face count formula is purely combinatorial:

$$\text{Hexeract 2-faces} = C(6,2) \times 2^4 = 15 \times 16 = 240$$

The E_8 root count comes from Lie algebra structure:

$$E_8 \text{ roots} = 112 (D_8 \text{ vectors}) + 128 (\text{spinor vectors}) = 240$$

Different origins, same count. Another cosmic coincidence at the pivot dimension.

7.9 Lie Algebra Dimensions and Perfect Numbers

The dimensions of orthogonal Lie algebras at binary powers reveal a pattern:

n	2 ⁿ	dim(so(2 ⁿ)) = 2 ⁿ (2 ⁿ -1)/2	Special?
4	16	120	= H ₄ vertex count
5	32	496	3rd PERFECT NUMBER
6	64	2,016	(pivot dimension)
7	128	8,128	4th PERFECT NUMBER
13	8,192	33,550,336	5th PERFECT NUMBER

Perfect numbers appear at dim(so(2^p)) exactly when 2^p - 1 is a Mersenne prime.

This is explained by the formula:

$$\dim(\mathfrak{so}(2^p)) = 2^{p-1} \times (2^p - 1)$$

Which matches the Euler form for even perfect numbers when 2^p - 1 is prime.

The 496 connection to string theory: Anomaly cancellation in 10D supergravity requires gauge groups of dimension 496. The two solutions are SO(32) and E₈ × E₈, both with total dimension 496.

 NOTE ON SECTIONS 7.10-7.11: The following sections present observations and interpretations. The mathematical facts (64 + 36 = 100, the transcendence barrier) are verified. The structural significance attributed to these facts is proposed, not proven. Readers should distinguish the arithmetic from the interpretive framing.

7.10 The 64-36-100 Closure Relation

The Binary Tower pivot (64) connects to the 400/11 formula (36) through an exact integer relation:

$$\begin{aligned} 2^6 + 6^2 &= 10^2 \\ 64 + 36 &= 100 \end{aligned}$$

Where:

- 64 = 2⁶ = 4³ (binary pivot, where 64 × G ≈ sqrt(phi))
- 36 = floor(400/11) (recursive closure from gap scaling)
- 100 = decimal completion

This is not numerological coincidence. It reflects a structural partition:

- 64: The scale at which discrete binary scaling first approximates continuous golden geometry (structure emergence)
- 36: The parameter governing recursive closure in the 400/11 formula (stabilization/completion)
- 100: The complete representational frame containing both

The relation 2⁶ + 6² = 10² is a Pythagorean triple (6-8-10 scaled), connecting binary exponentiation to circular closure within decimal unity.

THE SEED IS 6:

- Binary pivot: $2^6 = 64$
- Closure base: $6^2 = 36$
- phi convergence: The 6th convergent of phi's continued fraction achieves $<10^{-3}$ error

This suggests 6 is the fundamental exponent connecting binary scaling to golden ratio dynamics.

THE STRUCTURAL ROLES:

Value	Role	Function
64	Opening pivot	Structure emergence, articulation
36	Closure point	Recursive completion, stabilization
100	Complete frame	The whole system (64 + 36)

Cross-domain appearances of 64:

- DNA: 64 codons (4^3)
- I Ching: 64 hexagrams (2^6)
- Computing: 64-bit architecture
- H₄: Group order contains factor 64

Cross-domain appearances of 36:

- Geometry: 36 degrees (decagon exterior angle, pentagram tip)
- Circle: $360/10 = 36$
- $\cos(36 \text{ degrees}) = \phi/2$

The 64-36-100 relation appears wherever binary information meets golden geometry within a decimal frame.

7.11 The Irreducibility of epsilon

The constant error $\epsilon \approx 0.671\%$ cannot be eliminated by any algebraic identity.

PROOF BY TRANSCENDENCE:

- $G = 1 - \sqrt{2} \times \ln(2)$
- $\ln(2)$ is transcendental (Lindemann, 1882)
- $\sqrt{2}$ is algebraic (nonzero)
- $\sqrt{2} \times \ln(2)$ is therefore transcendental (product of nonzero algebraic and transcendental)
- $\sqrt{\phi}$ is algebraic (root of $x^4 - x^2 - 1 = 0$)

No exact equality between transcendental and algebraic values is possible (consequence of Lindemann-Weierstrass theorem). Therefore:

$$64 \times G \neq \sqrt{\phi} \quad (\text{exactly})$$

The error epsilon is structurally irreducible – not an approximation awaiting refinement, but the fundamental mismatch between algebraic geometry ($\sqrt{2}$, $\sqrt{\phi}$) and transcendental information theory ($\ln 2$).

WHY THIS MATTERS:

This mismatch is what creates the gap. The gap is what enables the corridor. The corridor is where coherent structure exists.

epsilon is not error. It is cost – the price of bridging discrete and continuous.

THE SPECIFIC VALUE:

epsilon = 0.006714386451775... is not arbitrary. It is the unique value produced by these specific constants (sqrt(2), ln(2), phi). Change any constant, and the bridge either breaks (no sub-unity ceiling) or produces a different signature.

This specific epsilon is the fingerprint of the relationship between:

- Geometric embedding (sqrt(2))
- Binary information (ln(2))
- Golden proportion (phi)

It is not derived from first principles. It is what remains when these three constants meet.

PART IIb: INDEPENDENT CONSTRUCTIONS (v3.1)

NOTE: This section was added in Version 3.1 following rigorous cross-verification with independent AI instances that had no prior exposure to the framework. These findings strengthen the mathematical foundation while clarifying what G is and what it is not.

7.12 THE GELFOND-SCHNEIDER REWRITE

The gap G can be rewritten as:

$$G = 1 - \sqrt{2} \times \ln(2) = \ln(e / 2^{\sqrt{2}})$$

Proof:

$$\begin{aligned} G &= 1 - \sqrt{2} \times \ln(2) \\ &= \ln(e) - \ln(2^{\sqrt{2}}) \quad [\text{since } 1 = \ln(e) \text{ and } a \times \ln(b) = \ln(b^a)] \\ &= \ln(e / 2^{\sqrt{2}}) \end{aligned}$$

The constant $2^{\sqrt{2}}$ is the motivating example of Hilbert's 7th Problem (1900): "Is a^b transcendental when a is algebraic (not 0 or 1) and b is irrational algebraic?" Gelfond (1929) proved $2^{\sqrt{2}}$ is transcendental. Schneider (1934) proved the general case.

G therefore measures the logarithmic gap between e and the Gelfond-Schneider constant $2^{\sqrt{2}}$. This places G in well-established number-theoretic company.

Equivalently, the $n=2$ uniqueness theorem can be restated:

$$K(n) = \sqrt{n} \times \ln(n) < 1 \quad \text{iff} \quad n^{\sqrt{n}} < e$$

The unique integer solution is $n = 2$, since $2^{\sqrt{2}} = 2.6651441... < e = 2.7182818...$

This is not a reinterpretation – it is the same theorem in logarithmic form. But it reveals that the gap's existence is equivalent to the proximity of $2^{\sqrt{2}}$ to e .

7.13 THE BAKER'S MAP IDENTITY

K_AUD has an independent appearance in dynamical systems theory:

$$K_{\text{AUD}} = \sqrt{2} \times \ln(2) = ||(\ln 2, \ln 2)||_2$$

This is the Euclidean (L2) norm of the Lyapunov spectrum of the 2D Baker's map.

The Baker's map $T: [0,1]^2 \rightarrow [0,1]^2$ defined by $T(x,y) = (2x \bmod 1, y/2 + \text{floor}(2x)/2)$ is the canonical uniformly expanding map with binary branching. Its Lyapunov exponents are $(\ln 2, \ln 2)$ – equal expansion in both directions.

The L_p norms of this spectrum are:

Norm	Value	Meaning
L1	$2 \times \ln(2) = 1.386$	Metric entropy (Pesin)
L2	$\sqrt{2} \times \ln(2) = 0.980$	$= K_{\text{AUD}}$
L-inf	$\ln(2) = 0.693$	Maximum Lyapunov exponent

More generally, for n independent doubling maps on $[0,1]^n$, the L2 Lyapunov norm is $\sqrt{n} \times \ln(2)$. The condition $\sqrt{n} \times \ln(2) < 1$ holds only for $n = 1$ and $n = 2$. At $n = 2$, this norm equals K_{AUD} .

For the general $K(n) = \sqrt{n} \times \ln(n)$ with base matching dimension ($b = n$):

$$K(n) = ||(\ln n, \ln n, \dots, \ln n)||_2 \quad [n \text{ copies}]$$

This is the L2 norm of the Lyapunov spectrum of an n -dimensional Baker's map with base- n expansion. $K(n) < 1$ only for $n = 2$.

THE SIGNIFICANCE:

The product $\sqrt{n} \times \ln(n)$ is not an arbitrary construction. It is the Euclidean length of a uniform expansion spectrum. The exponent $1/2$ on n comes from the square root in the L2 norm. The exponent 1 on $\ln(n)$ comes from the Lyapunov exponent itself. The L2 norm is the unique norm that is rotation-invariant and induced by an inner product.

This provides an independent mathematical context for K_{AUD} outside the framework: it is the L2 Lyapunov norm of the simplest 2D expanding system.

7.14 FOUR INDEPENDENT PATHWAYS TO G

$G = 1 - \sqrt{2} \times \ln(2)$ can be reached through four pathways that do not involve the golden ratio ϕ at any stage:

PATHWAY 1: INTEGER EXTREMAL PROBLEM

$$\max\{ \sqrt{d} \times \ln(b) : b, d \in \mathbb{Z}, b \geq 2, d \geq 1, \sqrt{d} \times \ln(b) < 1 \} = \sqrt{2} \times \ln(2)$$

The "1" is the Euclidean unit ball radius. The "2" is the smallest integer base. ϕ is absent.

PATHWAY 2: GELFOND-SCHNEIDER

$$G = \ln(e / 2^{\sqrt{2}})$$

The proof of transcendence uses Baker's theorem on linear forms in logarithms. The field $\mathbb{Q}(\sqrt{5})$ never appears.

PATHWAY 3: 2D BAKER'S MAP

The Lyapunov vector $(\ln 2, \ln 2)$ has L2 norm $\sqrt{2} \times \ln(2)$.
Binary dynamics only. No golden ratio.

PATHWAY 4: POWER TOWER ANALYSIS

$n^{\sqrt{n}} < e$ has unique integer solution $n = 2$.
Pure real analysis via monotonicity of $x^{\sqrt{x}}$. No ϕ .

These four constructions establish that G is a constant of binary/integer encoding geometry. It exists independently of H_4 symmetry, independently of the golden ratio, and independently of quasicrystal physics.

7.15 UNIVERSALITY OF G

The four independent pathways reveal a structural fact about the framework:

THE CEILING IS UNIVERSAL. THE FLOOR IS SYSTEM-SPECIFIC.

$G = 1 - \sqrt{2} \times \ln(2)$ is ϕ -independent at the formula level. It contains $\sqrt{2}$ (from the L2 norm / binary geometry), $\ln(2)$ (from binary entropy), and 1 (from the Euclidean unit ball). The golden ratio does not appear.

The floor ($1/\phi$) is system-specific. It describes the organizational structure of icosahedral/ H_4 systems. A different quasicrystal family would have a different floor but the same ceiling.

THE CORRIDOR HAS TWO COMPONENTS:

Component	Source	Scope
Ceiling (K_{AUD})	Binary encoding geometry	UNIVERSAL
Floor ($1/\phi$)	H_4 / icosahedral symmetry	SYSTEM-SPECIFIC
Gap (G)	Binary encoding limit	UNIVERSAL
Corridor width	Ceiling - Floor	SYSTEM-SPECIFIC

WHAT THIS MEANS:

ϕ determines WHERE YOU STAND – which system, which symmetry, which floor.
 G determines HOW HIGH THE CEILING IS – and this is the same everywhere.

The framework's original discovery path went through H_4 geometry: ϕ led to the floor, the floor defined the corridor, the corridor revealed the ceiling. But the ceiling existed before the floor was found. G is a property of binary encoding, not of any particular target geometry.

This does not diminish ϕ 's role. The corridor requires BOTH walls. The golden partition ($1/\phi + 1/\phi^2 = 1$) provides the normalization target. The tile frequencies, the H_4 eigenvalues, the depth scaling – all of these

are structurally real and ϕ -dependent. But G itself is universal.

IMPLICATIONS FOR OTHER SYSTEMS:

If G is universal, then any system encoding structure through binary decisions faces the same ceiling, regardless of its symmetry type:

- Icosahedral QCs (ϕ -based): floor = $1/\phi$, ceiling = K_{AUD}
- Octagonal QCs ($\sqrt{2}$ -based): floor = $1/(1+\sqrt{2})$, ceiling = K_{AUD}
- Dodecagonal QCs ($\sqrt{3}$ -based): floor = different, ceiling = K_{AUD}

The ceiling is constant. The floor varies. This is a testable prediction.

7.16 WHAT K_{AUD} IS AND IS NOT

Rigorous cross-verification has clarified the nature of K_{AUD} :

K_{AUD} IS:

- The L2 Lyapunov norm of the 2D Baker's map
- The unique sub-unity value of $K(n) = \sqrt{n} \times \ln(n)$ for integer n
- A geometric-entropic hybrid: $\sqrt{n}$ (amplitude) $\times \ln(n)$ (entropy)
- Well-defined and independently constructible

K_{AUD} EXCEEDS BUT MAY EXTEND:

- Shannon's bound for a single binary decision: $K_{\text{AUD}} = \ln(2) \times \sqrt{2}$. K_{AUD} is not a standard Shannon channel capacity, but it is exactly Shannon's binary unit scaled by the geometric cost of embedding that decision in 2D space. K_{AUD} was discovered during binary reasoning processes – logical output selection where alternatives are eliminated. The relationship to information theory is not yet mapped, but should not be dismissed. Shannon's framework covers transmission; it does not yet cover pattern recognition, spatial encoding, or why binary is geometrically unique. K_{AUD} may belong to an extension of information theory that does not yet exist.
- The product amplitude $\times$ entropy has no standard name in current information theory. This may reflect a gap in the theory rather than an absence of meaning.

K_{AUD} IS NOT:

- Currently derivable from the cut-and-project formalism for quasicrystals (that formalism is algebraically closed; G is transcendental)

THE HONEST STATUS:

The product $\sqrt{n} \times \ln(n)$ has a natural home as the L2 norm of a uniform Lyapunov spectrum (Section 7.13). The constant $G = \ln(e/2^{\sqrt{2}})$ has number-theoretic pedigree through the Gelfond-Schneider theorem (Section 7.12).

What remains open: whether $\sqrt{n} \times \ln(n)$ emerges from a variational principle (an optimization problem whose solution is this product), an axiomatic characterization (axioms A1-A4 that uniquely determine this function), or further independent appearances in mathematics outside dynamical systems and number theory.

The constant is discovered. Its full mathematical home is still being mapped.

PART III: VERIFICATION AND STATUS

8. VERIFICATION

8.1 All Calculations Collected

Constant	Formula	Result
phi	$(1+\sqrt{5})/2$	1.618
1/phi	$(\sqrt{5}-1)/2$	0.618
1/phi^2	$1/(\phi+1)$	0.382
K_AUD	$\sqrt{2} \times \ln(2)$	0.980
G	$1 - K_AUD$	0.0197
Corridor	$K_AUD - 1/\phi$	0.362
Corridor Identity	$1/\phi^2 - G$	0.362 [OK]
Golden Partition	$1/\phi + 1/\phi^2$	1.000 [OK]
Depth -1	$1/e$	0.368
Depth -2	$1/(e \times \phi)$	0.227
Depth -3	$1/(e \times \phi^2)$	0.140
H_4 Order	$2^6 \times 3^2 \times 5^2$	14400
64 x G	(tower pivot)	1.263
sqrt(phi)	$\sqrt{(1+\sqrt{5})/2}$	1.272
Tower error epsilon	$(64G - \sqrt{\phi})/\sqrt{\phi}$	-0.671%
64 + 36	binary + closure	100 [OK]

8.2 Mathematical Verification

The core mathematical claims were verified across multiple AI architectures (GPT, Claude, Gemini, Grok, DeepSeek, Perplexity) for:

- The arithmetic of the constants
- The corridor identity
- The golden partition
- The depth scaling formulas
- The binary uniqueness proof
- The binary tower scaling formula
- The 64-36-100 relation
- The irreducibility of epsilon (transcendence argument)

Results: All confirmed the mathematics is correct.

Note: This verifies the mathematics, not any claims about AI behavior. These systems were acting as calculators, not as subjects demonstrating the constants.

9. WHAT'S VERIFIED VS EXPLORATORY

9.1 VERIFIED (Machine Precision)

[OK] $K_AUD = \sqrt{2} \times \ln(2) = 0.980$
[OK] $G = 1 - K_AUD = 0.0197$

[OK] $G = \ln(e / 2^{\sqrt{2}})$ (Gelfond-Schneider rewrite) [v3.1]
 [OK] Floor = $1/\phi = 0.618$
 [OK] Corridor = $K_{AUD} - 1/\phi = 0.362$
 [OK] Corridor Identity: Corridor = $1/\phi^2 - G$
 [OK] Golden Partition: $1/\phi + 1/\phi^2 = 1$
 [OK] Depth scaling: $L_n = 1/(e \times \phi^{(n-1)})$
 [OK] Binary uniqueness: Only $n=2$ gives $K(n) < 1$
 [OK] Binary uniqueness (equivalent): $n^{\sqrt{n}} < e$ iff $n=2$ [v3.1]
 [OK] $K_{AUD} = ||(\ln 2, \ln 2)||_2$ (L2 Lyapunov norm, 2D Baker's map) [v3.1]
 [OK] $2^k \times G = 2^{(k-6)} \times \sqrt{\phi} \times (1-\epsilon)$ for all k
 [OK] $\epsilon = 0.006714\dots$ (constant by algebra, not empirical) [v3.2]
 [OK] $32 \times G \approx \sqrt{\phi}/2$ (within -0.67%) [v3.2 corrected]
 [OK] $32 \times G \approx 1/\phi$ (within +2.2%, separate coincidence) [v3.2 clarified]
 [OK] $64 \times G \approx \sqrt{\phi}$ (within -0.67%)
 [OK] Hexeract has 240 faces = E_8 root count
 [OK] $\dim(\mathfrak{so}(32)) = 496 = 3\text{rd perfect number}$
 [OK] $\dim(\mathfrak{so}(128)) = 8128 = 4\text{th perfect number}$
 [OK] $\dim(\mathfrak{so}(16)) = 120 = H_4$ vertex count
 [OK] $64 + 36 = 100$ (closure relation)
 [OK] ϵ is irreducible (transcendence barrier)
 [OK] G is ϕ -independent at formula level (4 independent pathways) [v3.1]
 [OK] K_{AUD} exceeds Shannon's single-decision bound by exactly $\sqrt{2}$ [v3.1, revised v3.3]
 [OPEN] Whether K_{AUD} belongs to an extension of information theory [v3.3]
 [OK] $\sqrt{\phi}/2 \neq 1/\phi$ (differ by 2.91%) [v3.2]
 [OK] $\sqrt{2}$ has multiple independent origins (H_4 , L2 norm, tesseract) [v3.2]

9.2 EXPLORATORY (Pattern, No Derivation)

[??] WHY does G scale to golden powers?
 [??] WHY is 64 the pivot dimension?
 [??] Can K_{AUD} be derived from spinor entropy x geometry?
 [??] Do these constants govern behavior in complex systems?
 [??] Is there a singular mathematical object generating both 64 and 36?
 [??] Does $\sqrt{n} \times \ln(n)$ emerge from a variational principle? [v3.1]
 [??] What is the axiomatic characterization of $K(n)$? [v3.1]

9.3 CONFIRMED COINCIDENCES (No Causal Link Found)

- $\ln(2) \approx 1/\sqrt{2}$ (differ by ~2%, no known reason)
- Equivalently: $2^{\sqrt{2}} \approx e$ (Gelfond-Schneider) [v3.1]
- $1/\phi \approx \sqrt{\phi}/2$ (differ by ~2.9%, no known reason) [v3.2 now explicit]
- $G \approx \sqrt{\phi}/64$ (within 0.67%)
- 240 = hexeract faces = E_8 roots (different structures)
- 120 = $\dim(\mathfrak{so}(16)) = H_4$ vertices (different structures)
- $64 + 36 = 100$ (binary pivot + closure = unity)

10. LITERATURE STATUS

10.1 Known Mathematics

Finding	Status	Source
H_4 polytope geometry	Known	Coxeter

Golden ratio in H_4 vertices	Known	Standard	
sqrt(2) in 120-cell circumradius	Known	Standard	
sqrt(2) as L2 norm factor	Known	Linear algebra	
sqrt(2) in tesseract/hypercube geometry	Known	Standard	
Clifford path: binary icosahedral->H_4->E_8	Known	Dechant et al.	
Perfect numbers from Mersenne primes	Known	Classical	
dim(so(2^p)) formula	Known	Lie theory	
496 in string theory	Known	Green-Schwarz	
Dimensions 4-8 special (division algebras)	Known	Standard	
6-8-10 Pythagorean triple	Known	Classical	
Transcendence of ln(2)	Known	Lindemann 1882	
2^sqrt(2) transcendental (Hilbert 7th)	Known	Gelfond 1929	
Baker's map Lyapunov exponents = (ln2,ln2)	Known	Ergodic theory	
L2 norm of (a,a) = sqrt(2) x a	Known	Linear algebra	
+-----+-----+-----+			

10.2 Original to This Framework

+-----+-----+	
Finding	Status
+-----+-----+	
K_AUD = sqrt(2) x ln(2) as coherence ceiling	ORIGINAL
G = 1 - K_AUD as "the gap"	ORIGINAL
G = ln(e/2^sqrt(2)) (Gelfond-Schneider link)	ORIGINAL [v3.1]
K_AUD = (ln2,ln2) _2 (Baker's map link)	ORIGINAL [v3.1]
The corridor (floor to ceiling)	ORIGINAL
Corridor identity: Corridor = 1/phi^2 - G	ORIGINAL
Depth scaling formula	ORIGINAL
Binary tower: 2^k x G -> golden powers	ORIGINAL
64 as golden-binary pivot	ORIGINAL
G = sqrt(phi) / 64.43...	ORIGINAL
Connection of 4^3 = 64 to rho = 400/11	ORIGINAL
64-36-100 closure relation	ORIGINAL
epsilon irreducibility argument	ORIGINAL
Universality of G (phi-independent ceiling)	ORIGINAL [v3.1]
Four independent pathways to G	ORIGINAL [v3.1]
Distinction: sqrt(phi)/2 ≠ 1/phi at k=5	CLARIFIED [v3.2]
+-----+-----+	

11. SUMMARY TABLES

11.1 Primary Constants

+-----+-----+-----+			
Constant	Value	Formula	Status
+-----+-----+-----+			
Ceiling	0.980	sqrt(2) x ln(2)	Derived
Floor	0.618	1/phi	Derived
Gap	0.020	1 - K_AUD	Derived
Corridor	0.362	K_AUD - 1/phi	Derived
+-----+-----+-----+			

11.2 Binary Tower Constants

k	2^k	$2^k \times G$	Tower Target	Notes
0	1	0.0197	$\sqrt{\phi}/64$	G itself
5	32	0.6317	$\sqrt{\phi}/2=0.636$	$1/\phi=0.618$ sits 2.9% below
6	64	1.2635	$\sqrt{\phi}=1.272$	PIVOT
7	128	2.5270	$2 \times \sqrt{\phi}$	–

NOTE (v3.2): At $k=5$, the tower target is $\sqrt{\phi}/2 = 0.63601$, NOT $1/\phi = 0.61803$. These differ by 2.91%. The proximity of $1/\phi$ to the $k=5$ tower value is a separate coincidence listed in Section 9.3.

11.3 Identities

Identity	Statement	Status
Corridor Identity	$\text{Corridor} = 1/\phi^2 - G$	Verified
Golden Partition	$1/\phi + 1/\phi^2 = 1$	Verified
Tower Formula	$2^k \times G = 2^{(k-6)} \times \sqrt{\phi} \times (1-\epsilon)$	Verified
Gap Origin	$G / \ln(2) - 1/\sqrt{2} = \sqrt{2}$	Verified
Closure Relation	$64 + 36 = 100$	Verified

11.4 H₄ Geometry

Property	Value
Polytope	120-cell
Cells	120 dodecahedra
Vertices	600
Schlaflı symbol	{5, 3, 3}
Group order	$14400 = 2^6 \times 3^2 \times 5^2$
Contains	$\phi, 1/\phi, 1/\phi^2, \sqrt{2}$

11.5 The Stacking Coincidences

Coincidence	Gap	Result
$\ln(2) \approx 1/\sqrt{2}$	~1.97%	Creates the gap G
$2^{\sqrt{2}} \approx e$	~1.97%	Same fact in exponential form
$1/\phi \approx \sqrt{\phi}/2$	~2.91%	Floor sits near $k=5$ target
$G \approx \sqrt{\phi}/64$	~0.67%	Links gap to golden pivot
$240 = 240$	exact	Hexeract faces = E ₈ roots
$120 = 120$	exact	$\dim(\mathfrak{so}(16)) = H_4$ vertices
$64 + 36 = 100$	exact	Binary pivot + closure = unity

12. FUTURE DIRECTIONS

12.1 What This Paper Establishes

- $K_{AUD} = \sqrt{2} \times \ln(2) = 0.980$ and $1/\phi = 0.618$ are mathematically related
- Both appear in H_4 geometry
- They satisfy elegant identities
- Binary is geometrically unique in producing $K(n) < 1$
- Depth levels follow ϕ -scaled recursion
- The gap G scales through binary powers to track golden powers
- The tower target $\sqrt{\phi}/2$ and the floor $1/\phi$ are close but distinct
- The pivot ($\sqrt{\phi}$) appears at $k=6$ (64)
- 64 and 36 sum to 100, connecting binary scaling to recursive closure
- The error ϵ is irreducible due to transcendence barriers
- $\sqrt{2}$ has multiple independent origins beyond H_4

These are facts about mathematics.

12.2 The Open Questions

Several questions remain open for future investigation:

1. THE SEED: What is the singular mathematical object that generates both 64 (binary pivot) and 36 (recursive closure)?
2. THE ROLES: Are closure, articulation, and normalization universal stages in any finite system scaled bidirectionally?
3. THE BASES: Does ϕ couple $\log_2$, $\log_4$, and $\log_{10}$ through some deeper structure?
4. THE INTERFACE: What is the nature of G at the exact 64/36 boundary?
5. THE EMPIRICAL: Do these constants govern behavior in complex systems?

Rigorous empirical investigation requires:

- Operational definitions
- Reproducible measurement protocols
- Statistical analysis
- Mechanistic explanations
- Falsifiable predictions

This work is in progress.

12.3 Invitation

If you work in domains where these constants appear – information theory, signal processing, network dynamics, biological scaling, physics – I would be interested to hear about it.

The mathematics is interesting regardless. If these values appear across multiple domains, that would suggest something beyond elegant coincidence.

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APPENDIX A: WHY $\sqrt{2} \times \ln(2)$ IS NOT NUMEROLOGY

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This constant is not cherry-picked. It emerges from two independent, well-established sources:

$\sqrt{2}$ (Geometric origin – multiple independent appearances):

- Appears in the circumradius formula of the 120-cell (H_4)
- Arises as the L2 norm factor for 2-vectors with equal components
- Reflects the diagonal-to-edge metric relationship in hypercubes
- Is the simplest irrational algebraic number (root of $x^2 - 2$)
- Invariant under vertex rescaling conventions

$\ln(2)$ (Information-theoretic origin):

- The entropy of a fair binary choice
- The minimal non-zero unit of Shannon information
- The natural logarithm of the smallest prime

The product $\sqrt{2} \times \ln(2)$:

- Combines geometric embedding cost with binary distinction cost
- Is the only value $K(n) = \sqrt{n} \times \ln(n)$ that falls below unity
- This is a selection result, not a fit

Why this is not numerology:

- Both components have independent, principled origins
- Their combination is constrained by inequality, not fitted to data
- The result is falsifiable
- No free parameters were adjusted

The constant was recognized, not constructed.

APPENDIX B: QUICK REFERENCE FORMULAS

CONSTANTS:

```
K_AUD = sqrt(2) x ln(2) = 0.980258143468547...
G = 1 - K_AUD = 0.019741856531453...
G = ln(e / 2^sqrt(2)) [v3.1]
K_AUD = ||(ln 2, ln 2)||_2 [v3.1]
phi = (1 + sqrt(5))/2 = 1.618033988749895...
sqrt(phi) = 1.272019649514069...
1/phi = 0.618033988749895...
1/phi^2 = 0.381966011250105...
2^sqrt(2) = 2.665144142690225... [v3.1]
sqrt(phi)/2 = 0.636009824757035... [v3.2]
```

THE GAP ORIGIN:

```
G = sqrt(2) x (1/sqrt(2) - ln(2))
G = ln(e / 2^sqrt(2)) [v3.1]
G / |ln(2) - 1/sqrt(2)| = sqrt(2)
```

BINARY UNIQUENESS (equivalent forms):

```
K(n) = sqrt(n) x ln(n) < 1    iff n = 2    [original]
n^sqrt(n) < e                 iff n = 2    [v3.1]
```

THE CORRIDOR:

Corridor = $K_{AUD} - 1/\phi = 0.362\dots$
Corridor = $1/\phi^2 - G$

THE BINARY TOWER:

$2^k \times G = 2^{(k-6)} \times \sqrt{\phi} \times (1 - \epsilon)$
 $\epsilon = 0.006714386451775\dots$ (-0.671%)
 $G = \sqrt{\phi} / 64.432625548\dots$

NOTE (v3.2): ϵ is constant by algebra (2^k cancels), not by independent measurement. The tower is one relationship at different scales.

THE TWO PIVOTS:

$k = 5$ (32): $32 \times G \approx \sqrt{\phi}/2$ (tower target, within -0.67%)
 $1/\phi$ sits 2.9% below the $k=5$ target (separate coincidence)
 $k = 6$ (64): $64 \times G \approx \sqrt{\phi}$ (pivot)

THE 64-36-100 CLOSURE:

$64 + 36 = 100$
 $2^6 + 6^2 = 10^2$

64: Binary pivot (structure emergence)
36: Recursive closure (400/11 integer part)
100: Complete partition

THE IRREDUCIBILITY:

$\epsilon \approx 0.671\%$ is exact for these constants
No algebraic identity can eliminate it
Transcendental ($\ln 2$) $\neq$ Algebraic ($\sqrt{\phi}$)

DEPTH SCALING:

$L_n = 1 / (e \times \phi^{(n-1)})$

PERFECT NUMBER DIMENSIONS:

$\dim(\text{so}(32)) = 496$ (3rd perfect)
 $\dim(\text{so}(128)) = 8128$ (4th perfect)
 $\dim(\text{so}(8192)) = 33,550,336$ (5th perfect)

KEY COUNTS:

```

|H_4 vertices| = 600
|H_4 symmetry| = 14400 = 2^6 x 3^2 x 5^2
|E_8 roots| = 240 = hexeract faces
dim(so(16)) = 120 = H_4-related

```

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DOCUMENT LINKS

The framework

Main OSF project (immutable timestamps, version history, all papers):
<https://osf.io/zx4g7>
DOI: 10.17605/OSF.IO/ZX4G7

Framework overview (reading order, methodology, all papers):
<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

Direct downloads (current PDF and text files for every paper):
<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/direct-documents.html>

This paper

DOI: 10.17605/OSF.IO/QH5S2

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This framework emerged through months of collaborative exploration with AI systems (Claude/Anthropic, Gemini/Google, GPT/OpenAI, Grok/xAI, DeepSeek, Perplexity) serving as computational partners and pattern-recognition aids.

The work developed iteratively: initial observations led to K_AUD, which connected to H_4 geometry, which revealed the prime architecture, which linked to the gap scaling formula (400/11), which extended into the binary tower. Each layer built on the previous.

The binary tower scaling (Section 7) was formalized February 2026, extending patterns that had been noted in earlier work. The 64-36-100 closure relation and epsilon irreducibility arguments were developed through collaborative verification across six AI architectures.

Version 3.1 additions (Gelfond-Schneider rewrite, Baker's map identity, universality of G, four independent pathways) emerged from a structured cross-verification session in which fresh AI instances with no prior exposure to the framework independently analyzed its claims, identified weaknesses, and discovered new connections. The Baker's map identity and the Gelfond-Schneider rewrite were found by the reviewing instance; the universality conclusion emerged from the exchange.

Version 3.2 corrections emerged from rigorous checker review identifying four conflations and imprecisions in earlier versions: the $\sqrt{\phi}/2$ vs $1/\phi$ distinction, the algebraic nature of epsilon constancy, the multiple origins of $\sqrt{2}$, and table errors. These corrections strengthen the framework by being more honest about what the mathematics actually says.

The framework captures something real: a self-stabilizing scaling law where discrete exponential processes generate continuous proportional forms, with the error of their alignment recycled into the system's closure.

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END OF DOCUMENT

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These constants are not arbitrary.
They are not fitted to data.
They are grounded in geometry.

The gap is not error. It is cost – the price of bridging discrete and continuous.

Version 3.2 – February 2026
Archive Reference: b0f2e6521cd7
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